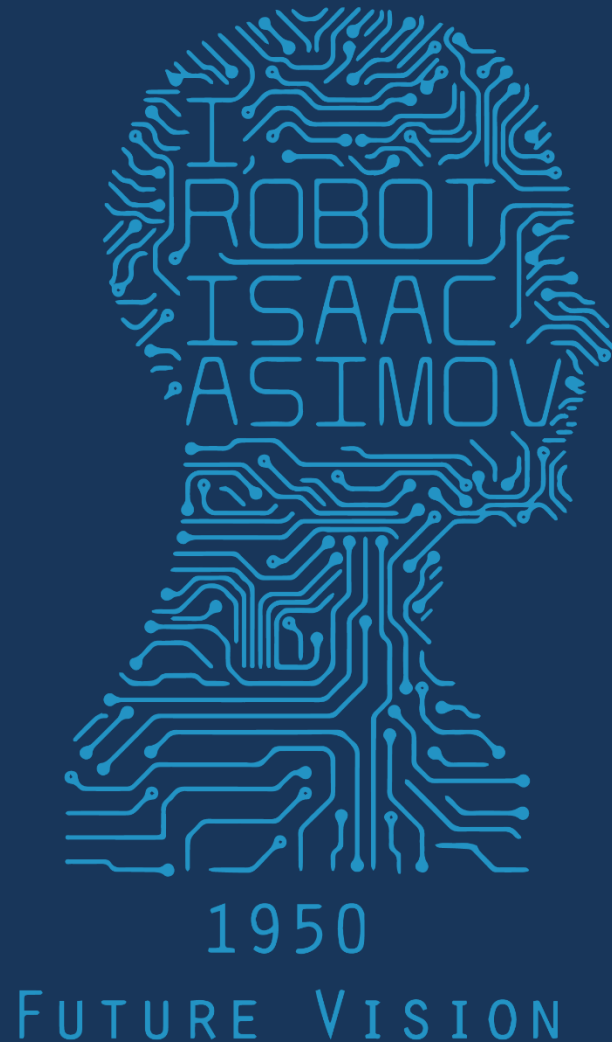


Computer Vision and Pattern Recognition



计算机视觉与 模式识别

Edge Detection

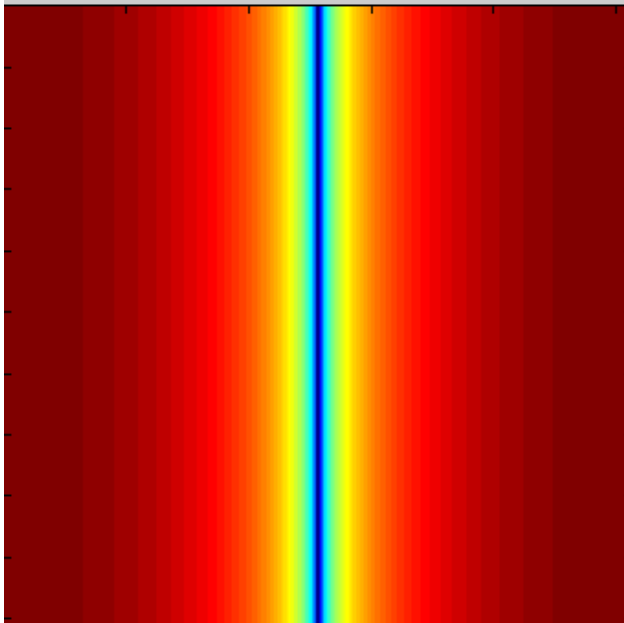
Review of Filtering

- Filtering in frequency domain
 - Can be faster than filtering in spatial domain (for large filters)
 - Can help understand effect of filter
 - Algorithm:
 1. Convert image and filter to fft (fft2 in matlab)
 2. Pointwise-multiply ffts
 3. Convert result to spatial domain with ifft2

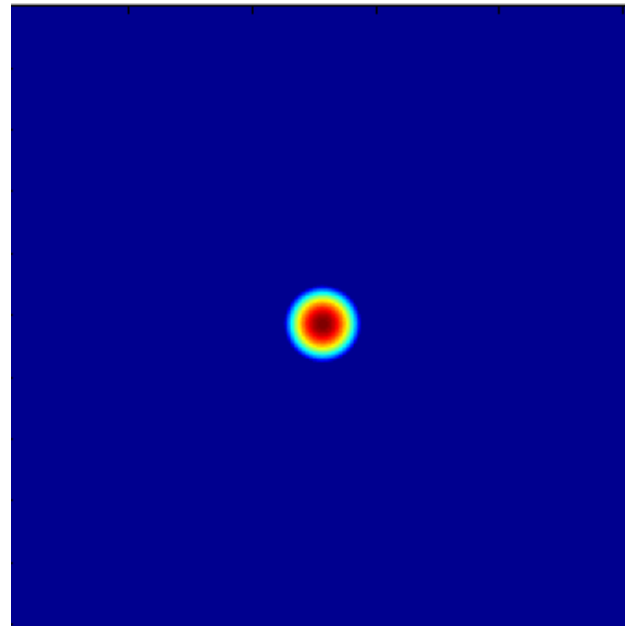
Review of Filtering

- Linear filters for basic processing
 - Edge filter (high-pass)
 - Gaussian filter (low-pass)

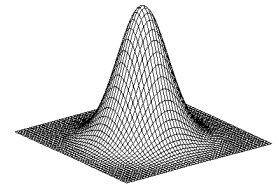
$[-1 \ 1]$



FFT of Gradient Filter



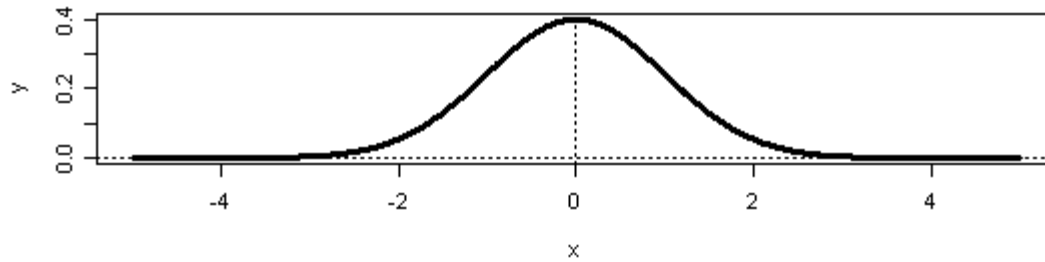
FFT of Gaussian



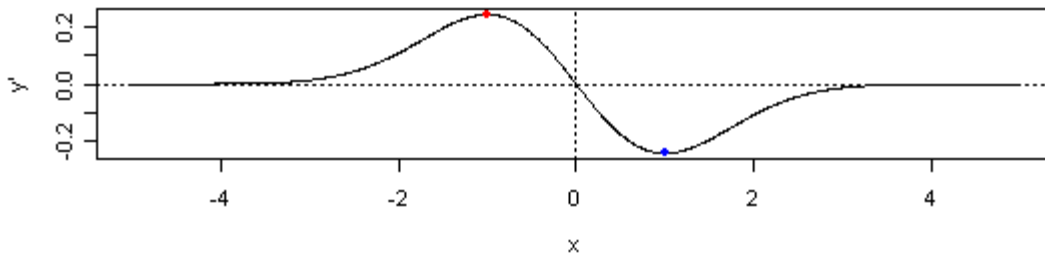
Gaussian

More Useful Filters

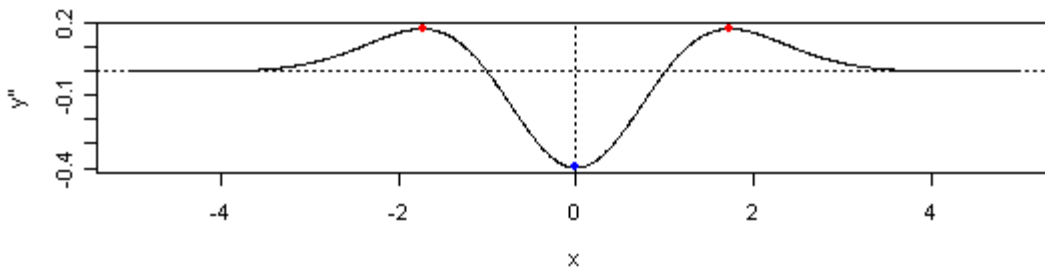
Single Gaussian



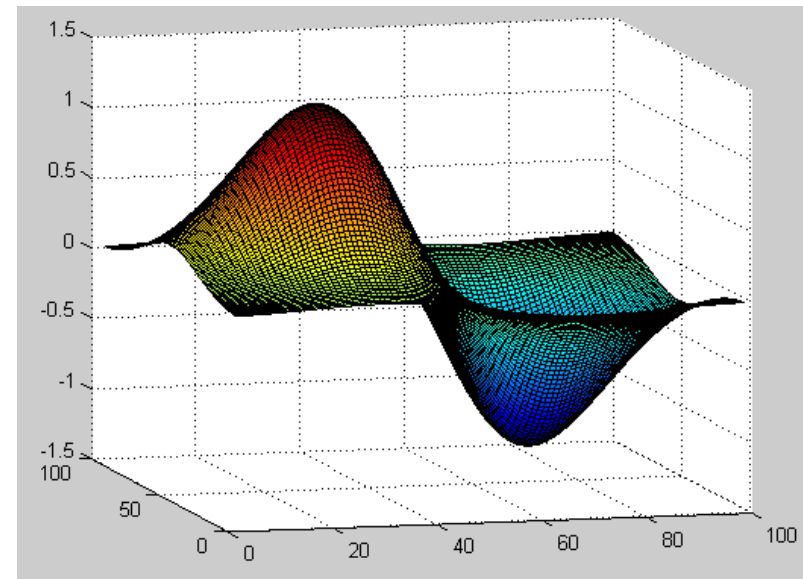
1st Derivative



2nd Derivative (Laplacian of Gaussian)

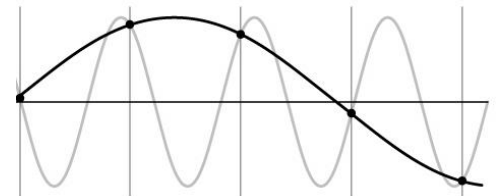
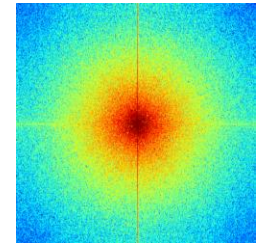
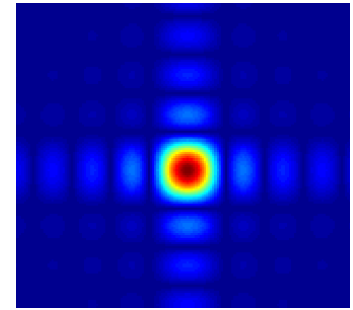
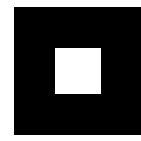


1st Derivative of Gaussian



Things to Remember

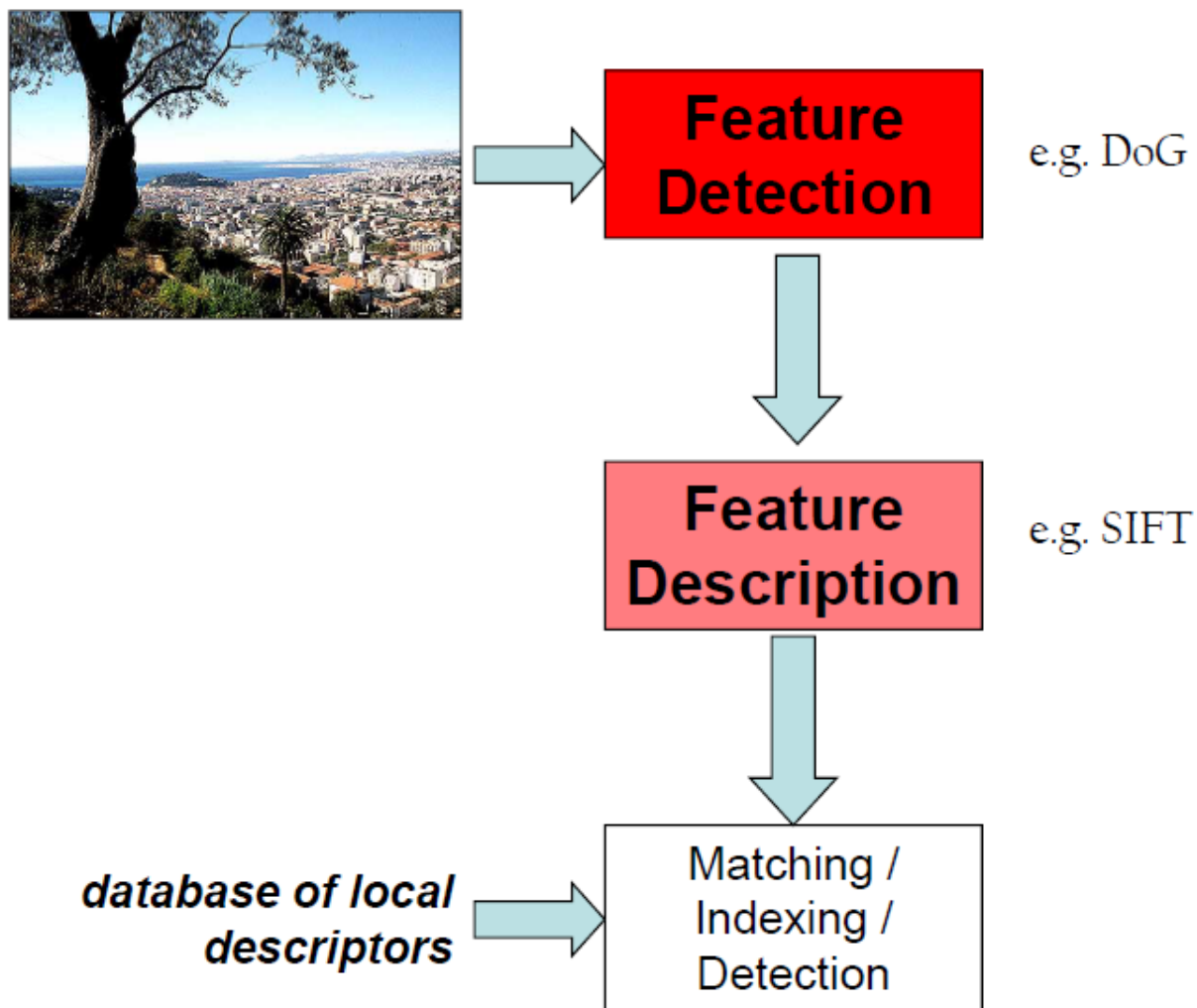
- Sometimes it makes sense to think of images and filtering in the frequency domain
 - Fourier analysis
- Can be faster to filter using FFT for large images
 - $N \log N$ vs. N^2 for auto-correlation
- Images are mostly smooth
 - Basis for compression
- Remember to low-pass before sampling
 - Otherwise you create aliasing



Previous Lectures

- We've now touched on the first three chapters of Szeliski.
 - 1. Introduction
 - 2. Image Formation
 - 3. Image Processing
- Now we're moving on to
 - 4. Feature Detection and Matching

The big picture...



EDGE / BOUNDARY DETECTION

Szeliski 4.2

Many slides from James Hays, Lana Lazebnik, Steve Seitz, David Forsyth, David Lowe, Fei-Fei Li, and Derek Hoiem



未来媒体研究中心
CENTER FOR FUTURE MEDIA



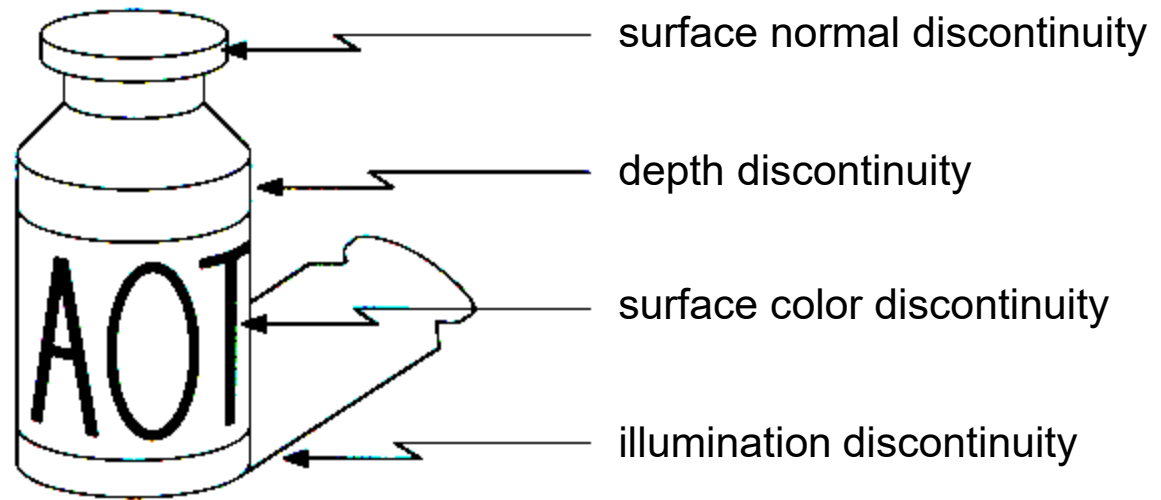
电子科技大学
University of Electronic Science and Technology of China

Edge detection

- **Goal:** Identify visual changes (discontinuities) in an image.
- Intuitively, semantic information is encoded in edges.
- What are some 'causes' of visual edges?



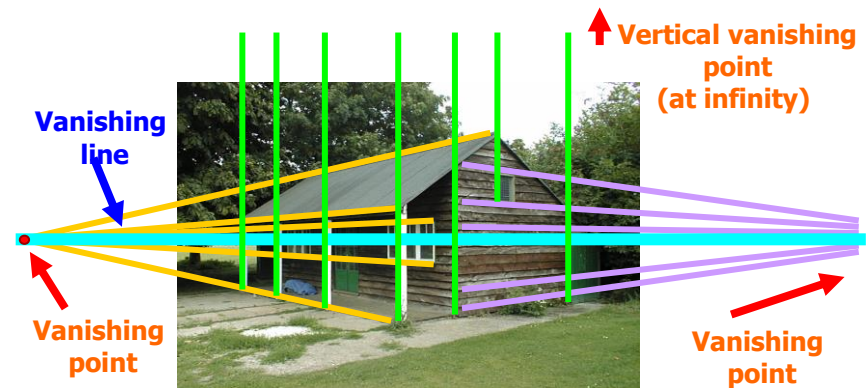
Origin of Edges



- Edges are caused by a variety of factors

Why do we care about edges?

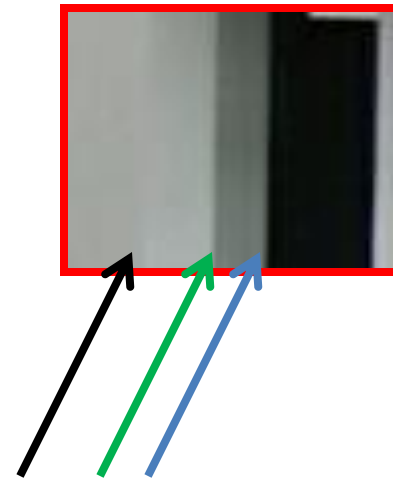
- Extract information
 - Recognize objects
- Help recover geometry and viewpoint



Closeup of edges



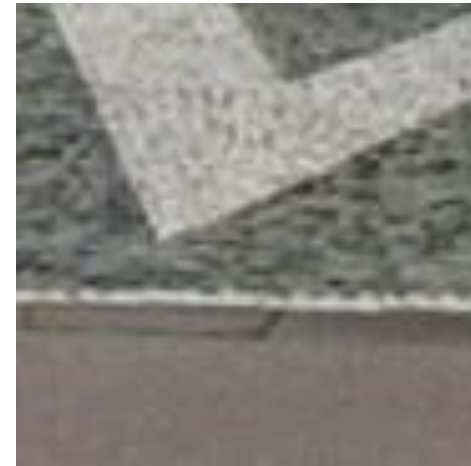
Closeup of edges



Closeup of edges



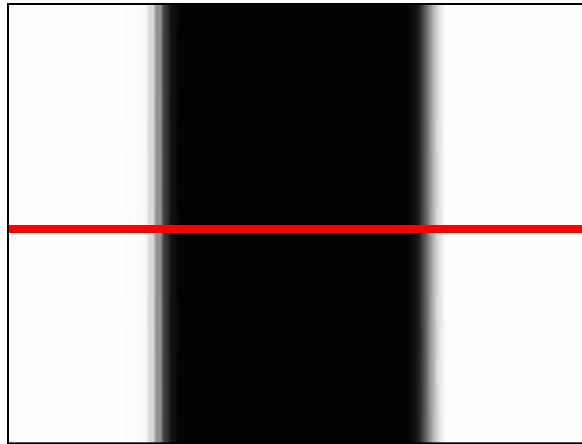
Closeup of edges



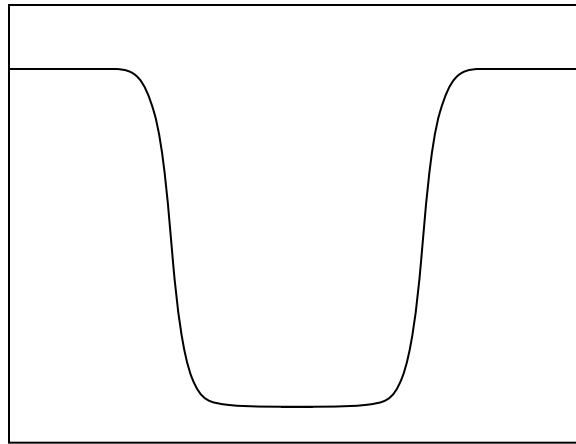
Characterizing edges

- An edge is **a place of rapid change in the image intensity function**

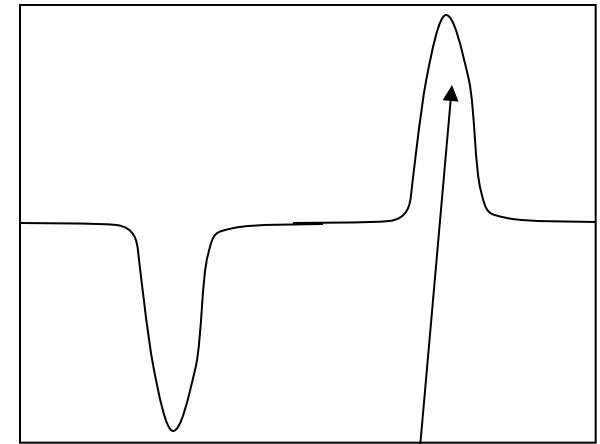
image



intensity function
(along horizontal scanline)

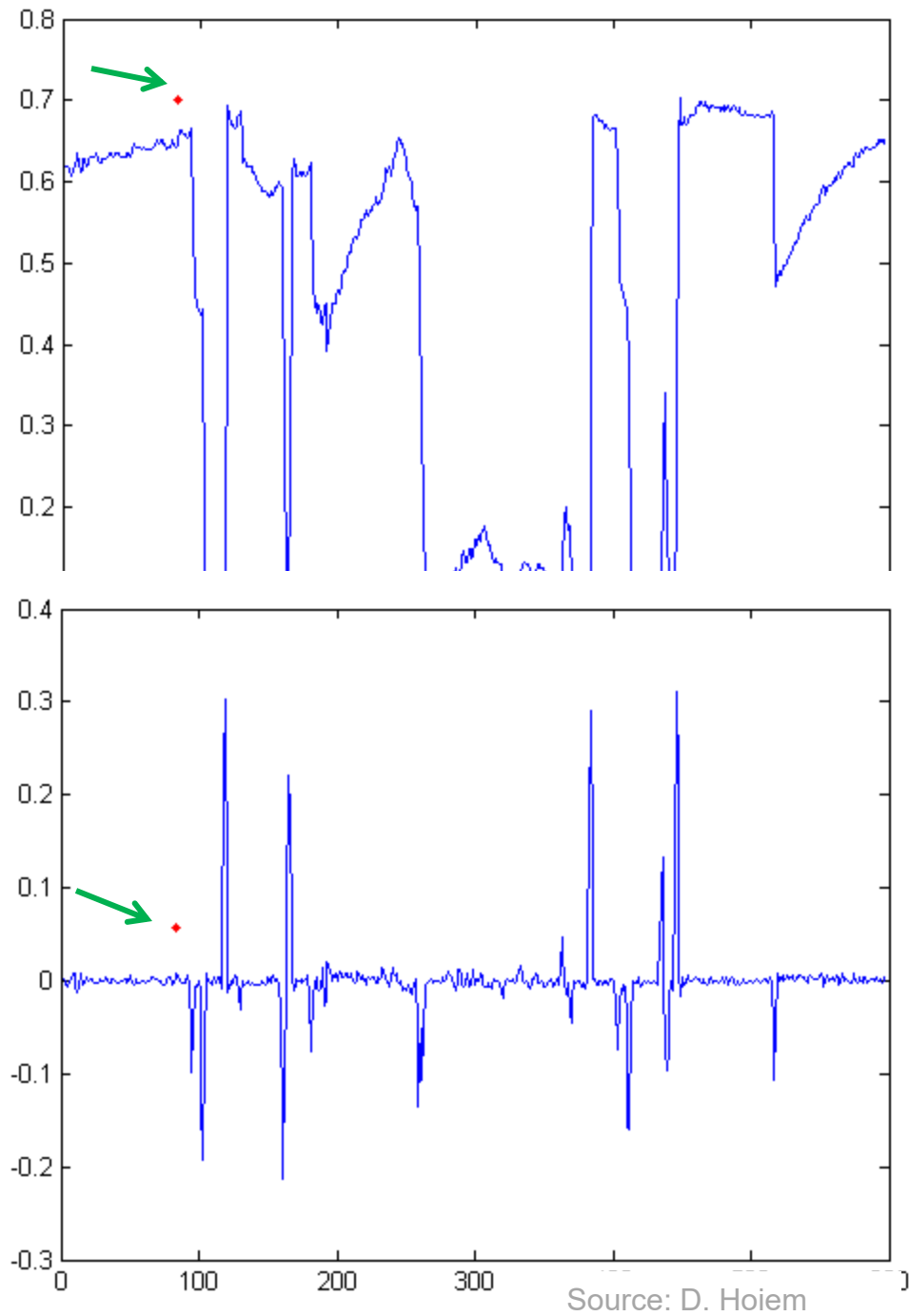
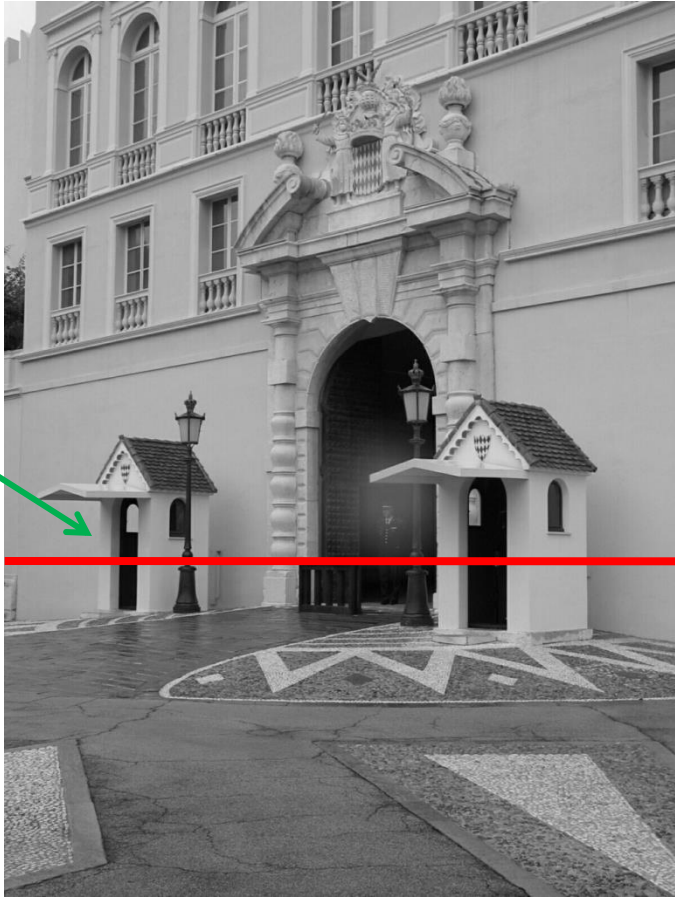


first derivative

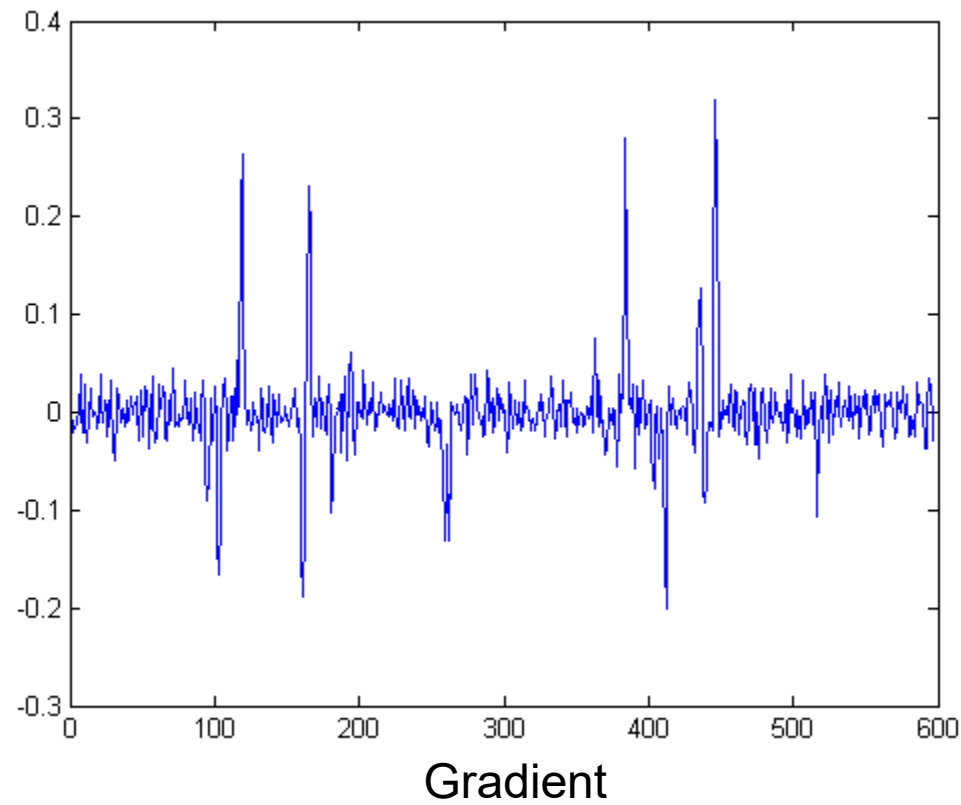
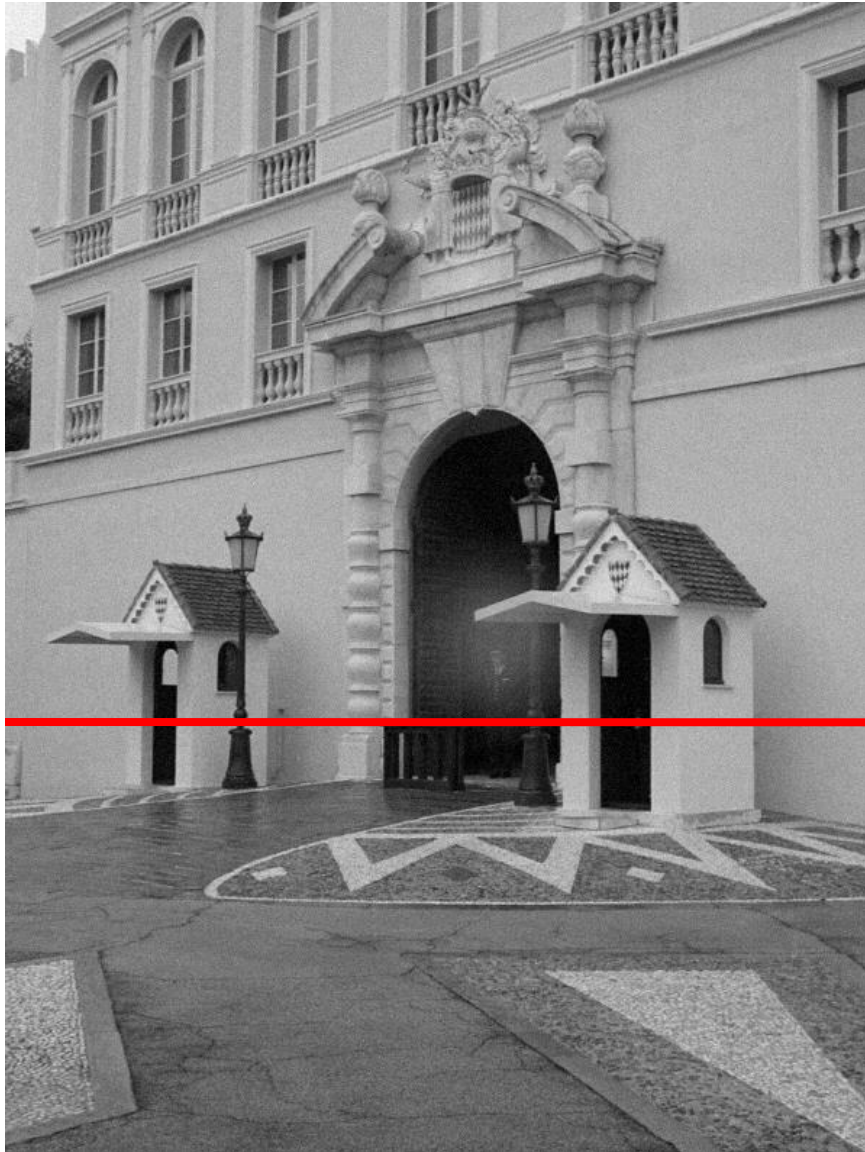


edges correspond to
extrema of derivative

Intensity profile

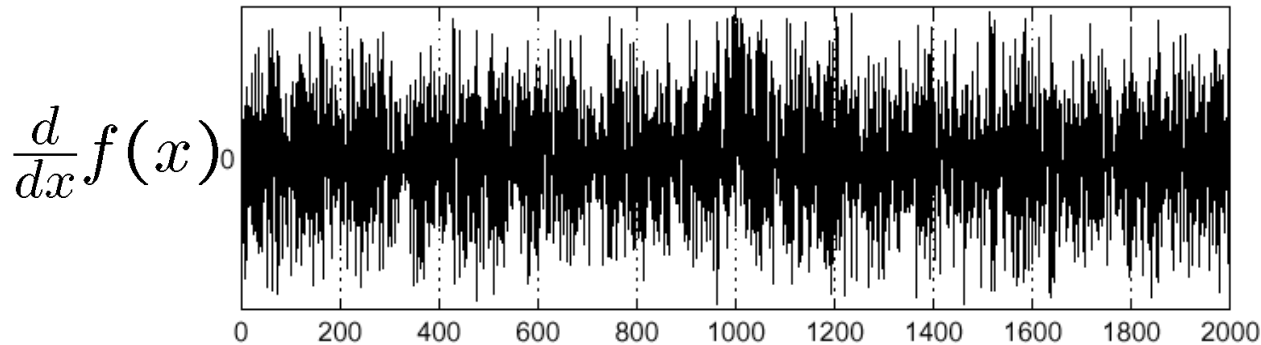
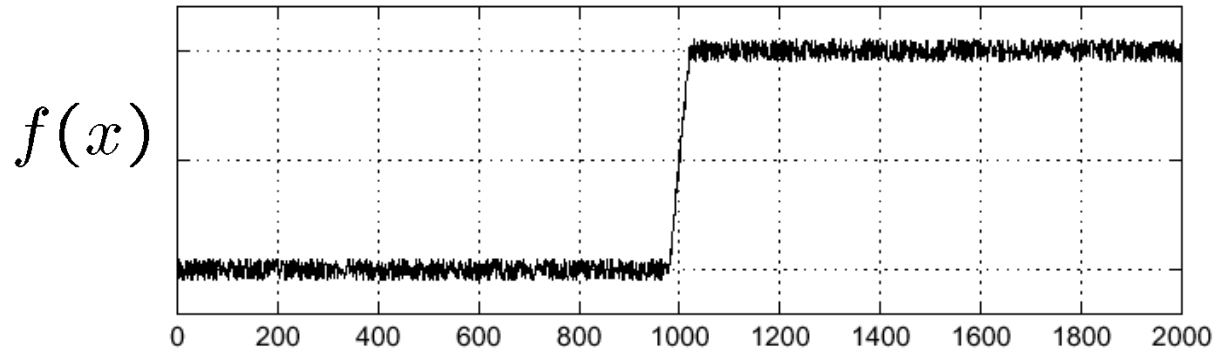


With a little Gaussian noise



Effects of noise

- Consider a single row or column of the image
 - Plotting intensity as a function of position gives a signal

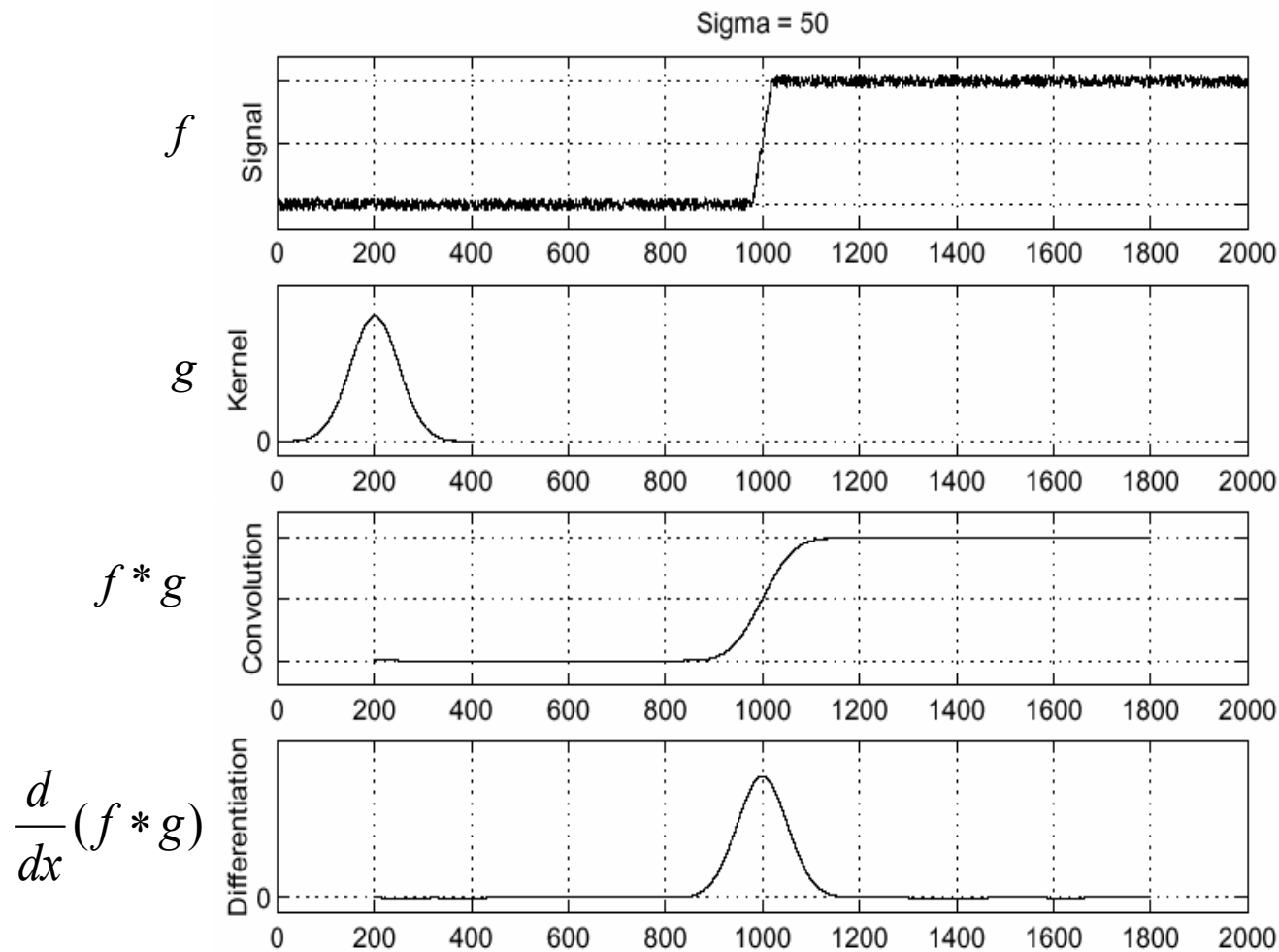


Where is the edge?

Effects of noise

- Difference filters respond strongly to noise
 - Image noise results in pixels that look very different from their neighbors
 - Generally, the larger the noise the stronger the response
- What can we do about it?

Solution: smooth first



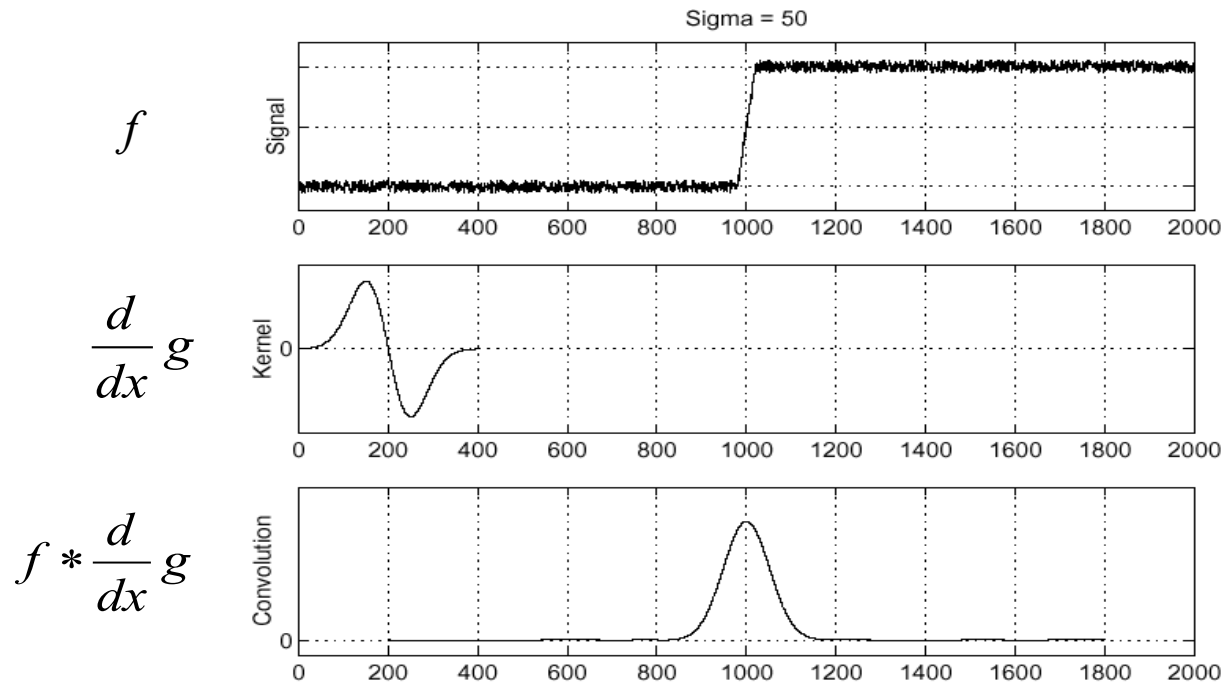
- To find edges, look for peaks in $\frac{d}{dx}(f * g)$

Derivative theorem of convolution

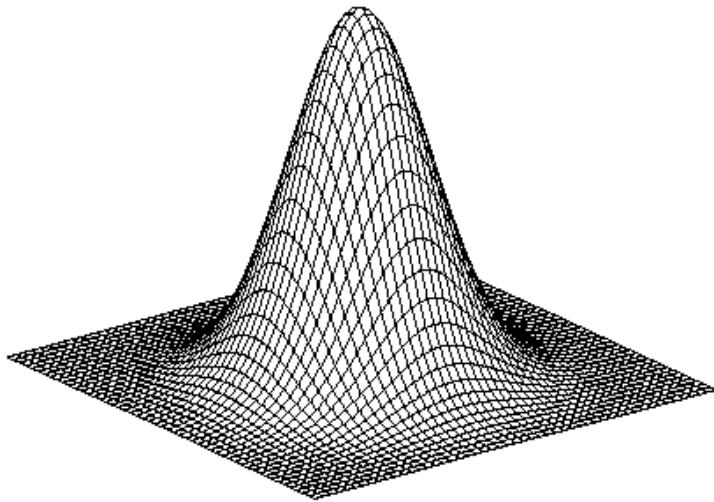
- Convolution is differentiable:

$$\frac{d}{dx}(f * g) = f * \frac{d}{dx}g$$

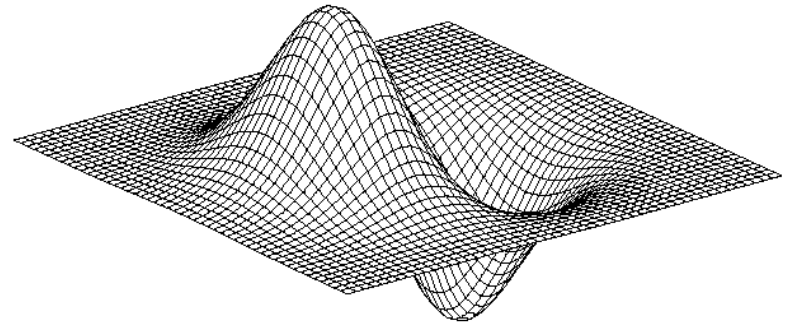
- This saves us one operation:



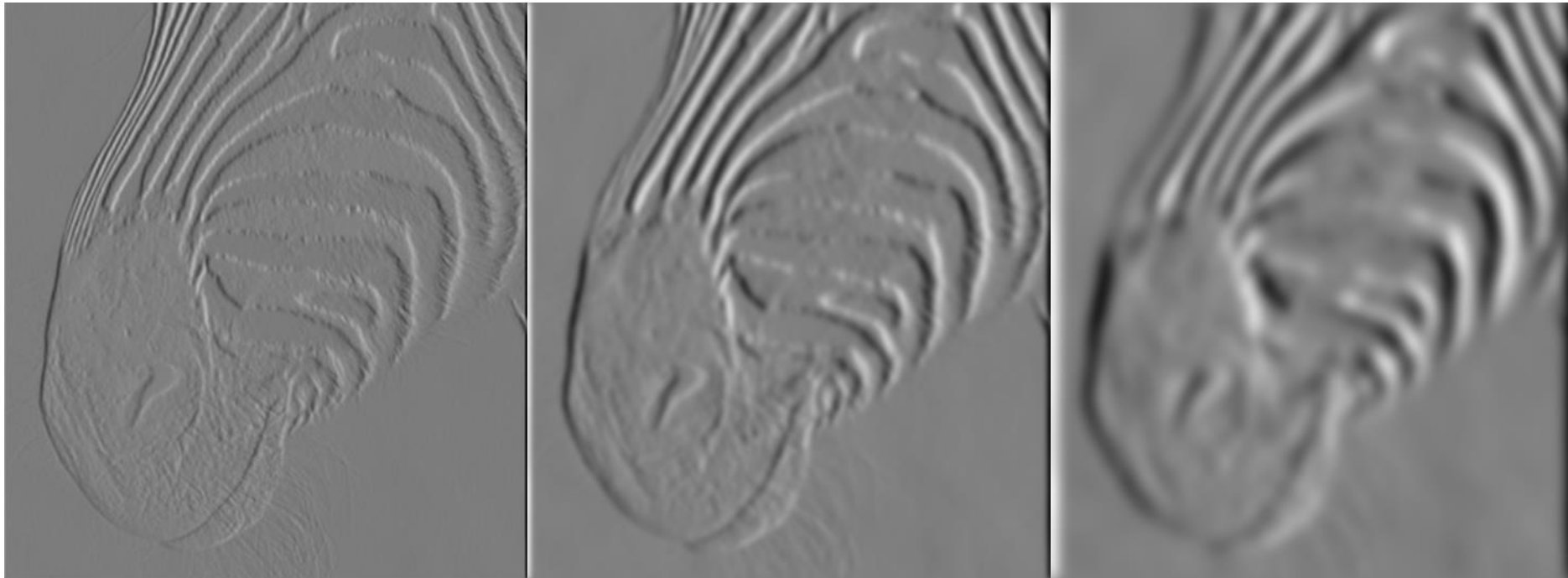
Derivative of 2D Gaussian filter



$$* [1 \ -1] =$$



Tradeoff between smoothing and localization



1 pixel

3 pixels

7 pixels

- Smoothed derivative removes noise, but blurs edge. Also finds edges at different “scales”.

Think-Pair-Share

What is a good edge detector?

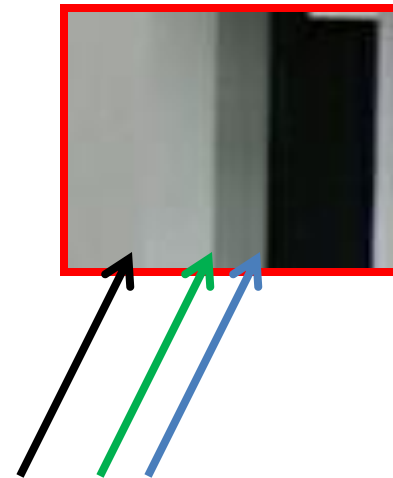
Do we lose information when we look at edges?

Are edges 'complete' as a representation of images?

Designing an edge detector

- Criteria for a good edge detector:
 - **Good detection:** the optimal detector should find all real edges, ignoring noise or other artifacts
 - **Good localization**
 - the edges detected must be as close as possible to the true edges
 - the detector must return one point only for each true edge point
- Cues of edge detection
 - Differences in color, intensity, or texture across the boundary
 - Continuity and closure
 - High-level knowledge

Closeup of edges



Elder – Are Edges Incomplete? 1999

Edge 'code':

- position,
- gradient magnitude,
- gradient direction,
- blur.

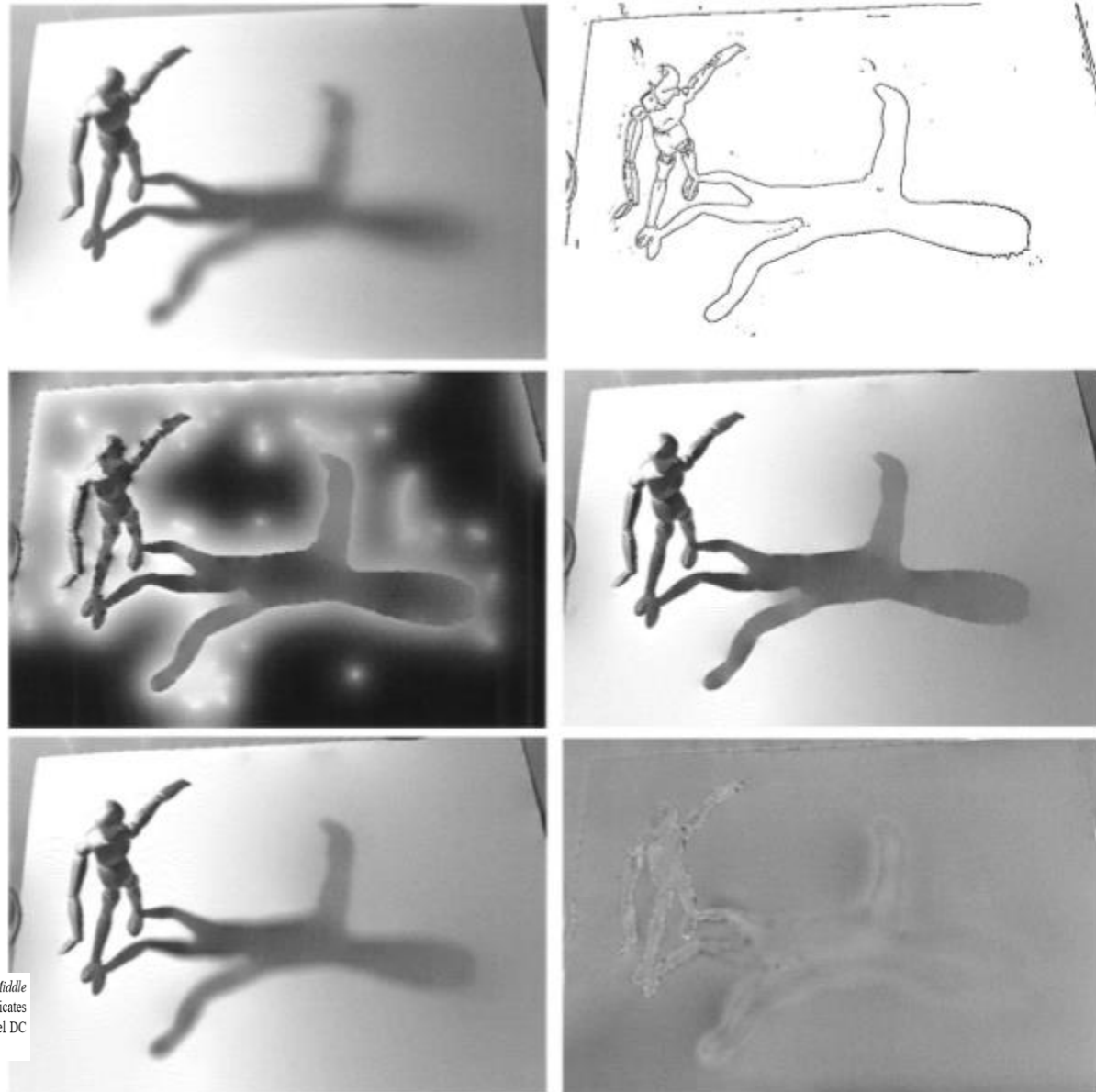


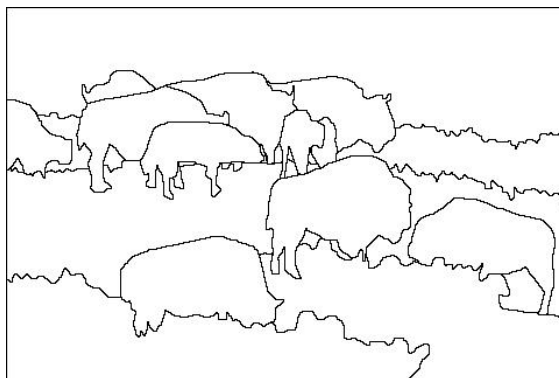
Figure 8. Top left: Original image. Top right: Detected edge locations. Middle left: Intermediate solution to the heat equation. Middle right: Reconstructed luminance function. Bottom left: Reblurred result. Bottom right: Error map (reblurred result—original). Bright indicates overestimation of intensity, dark indicates underestimation. Edge density is 1.7%. RMS error is 10.1 grey levels, with a 3.9 grey level DC component, and an estimated 1.6 grey levels due to noise removal.

Where do humans see boundaries?

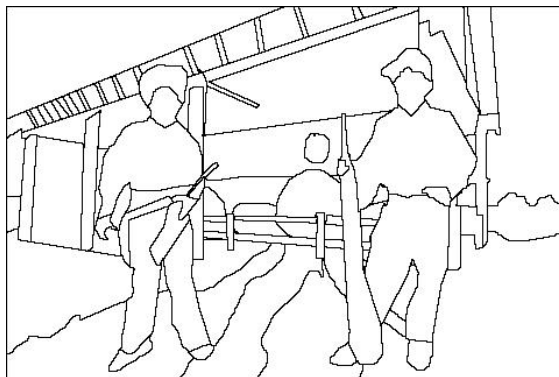
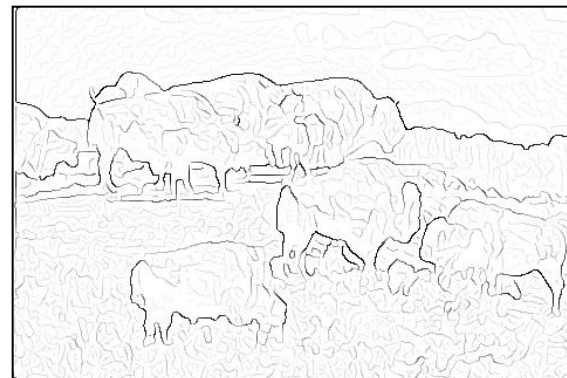
image



human segmentation



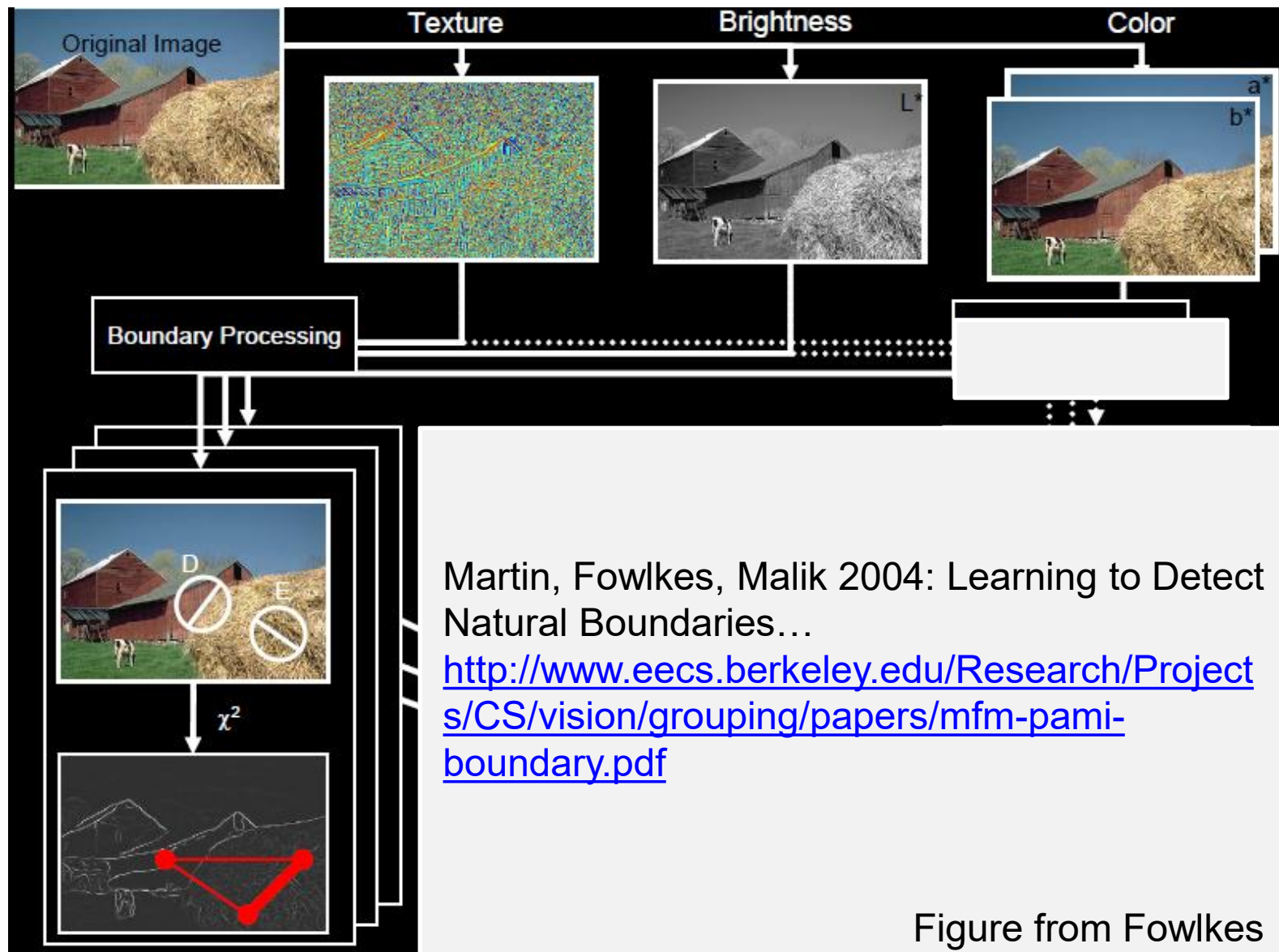
gradient magnitude



- Berkeley segmentation database:

<http://www.eecs.berkeley.edu/Research/Projects/CS/vision/grouping/segbench/>

pB boundary detector



Martin, Fowlkes, Malik 2004: Learning to Detect Natural Boundaries...

<http://www.eecs.berkeley.edu/Research/Projects/CS/vision/grouping/papers/mfm-pami-boundary.pdf>

Figure from Fowlkes



Brightness

Color

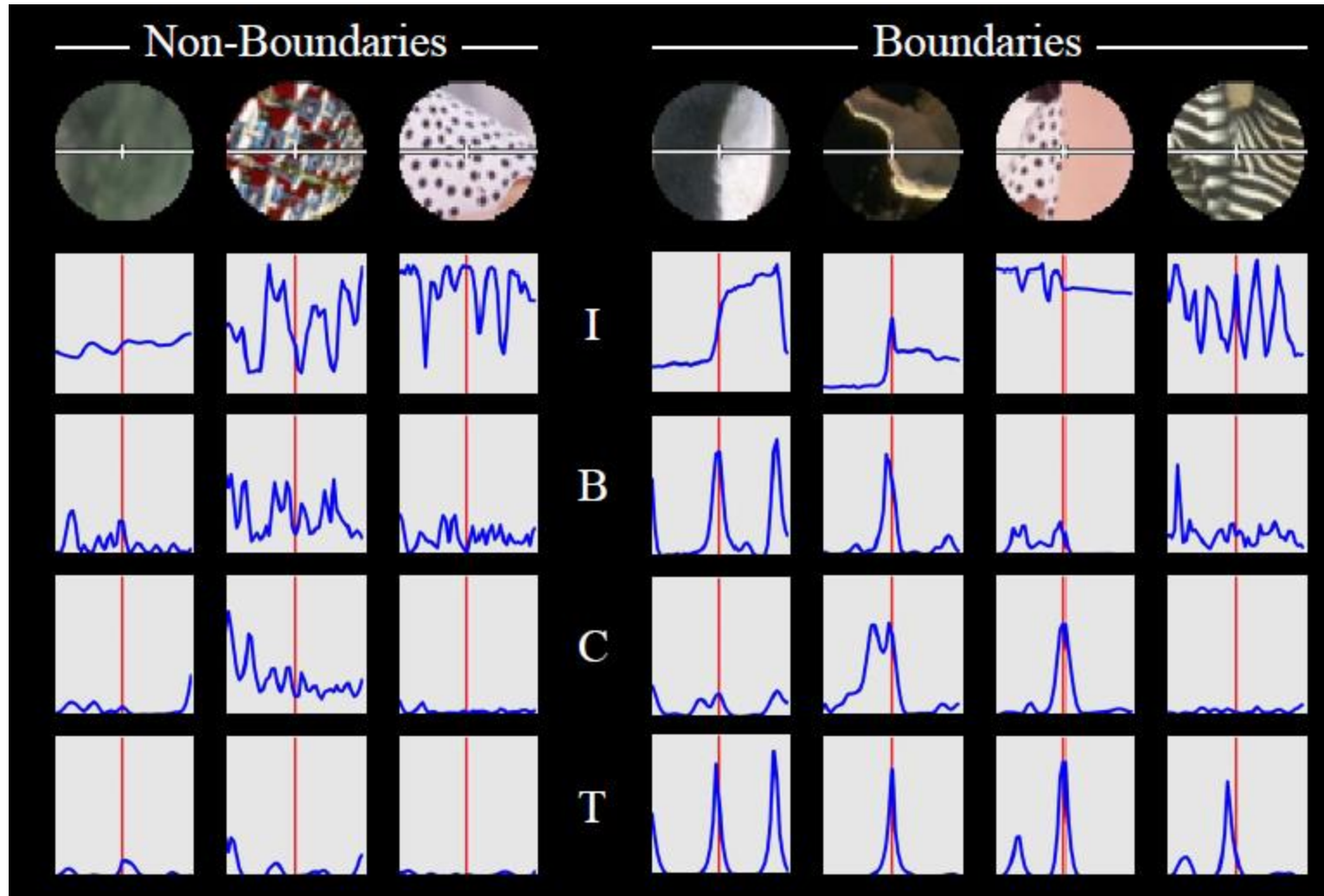
Texture

Combined

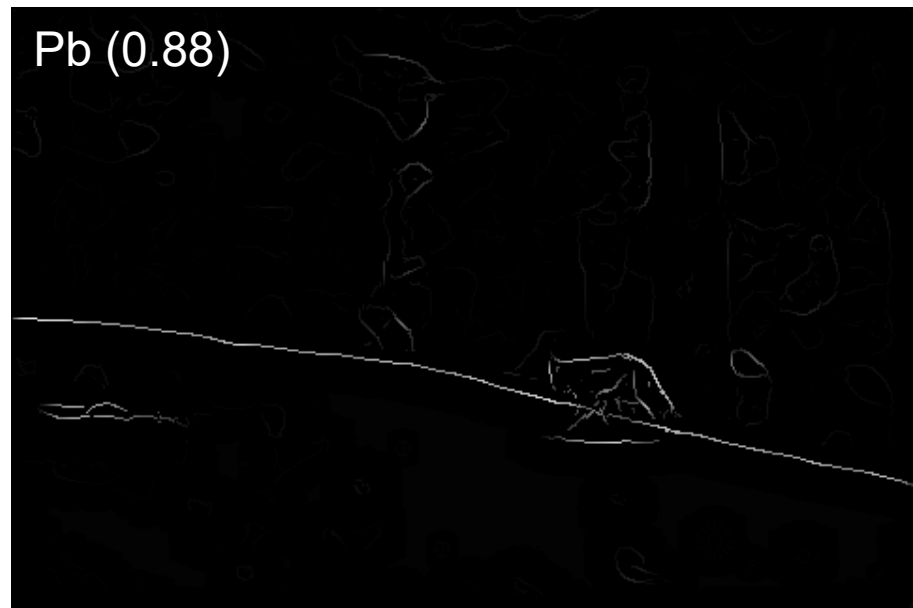
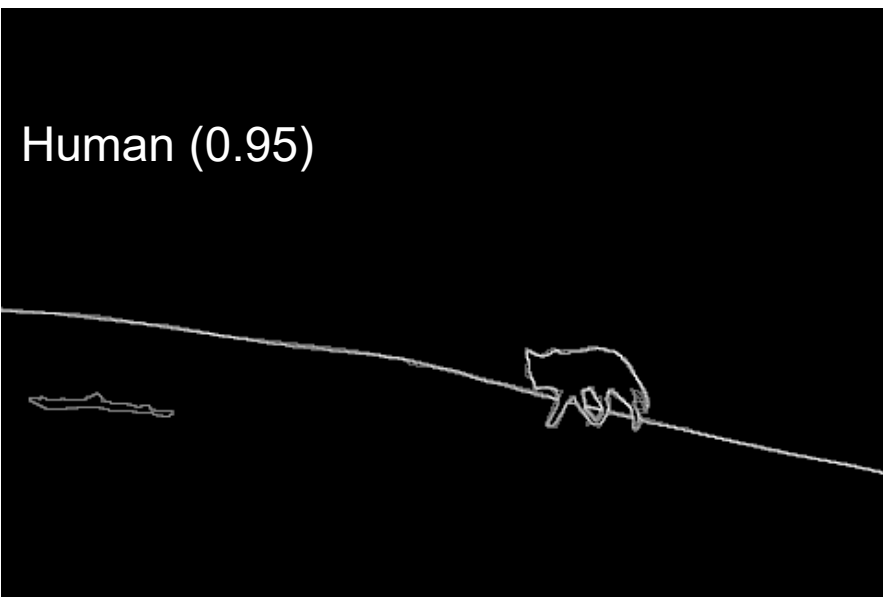
Human



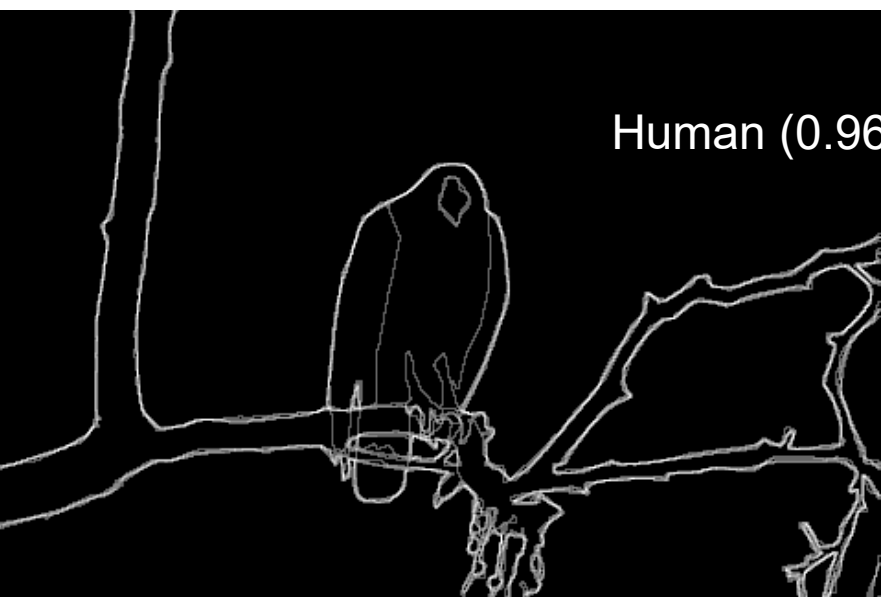
pB Boundary Detector

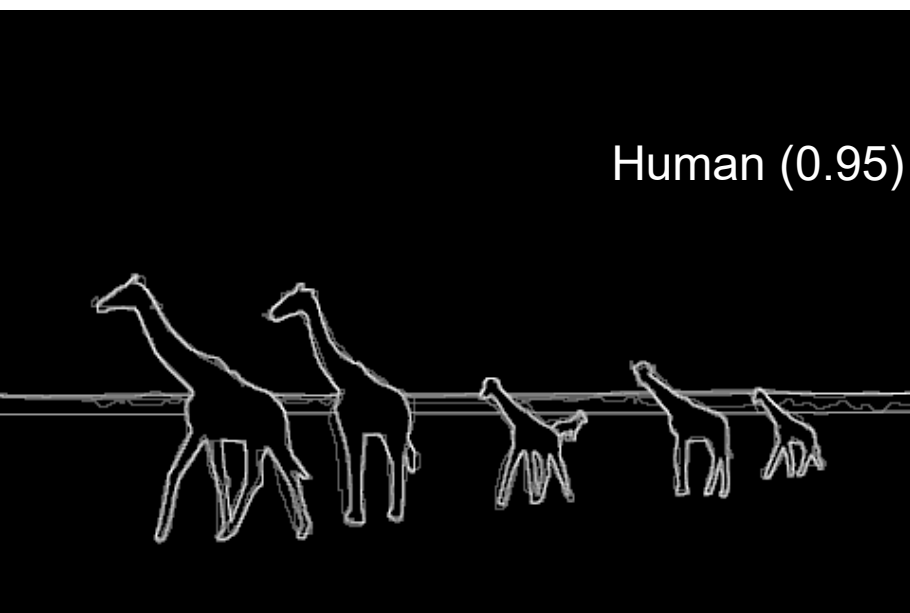


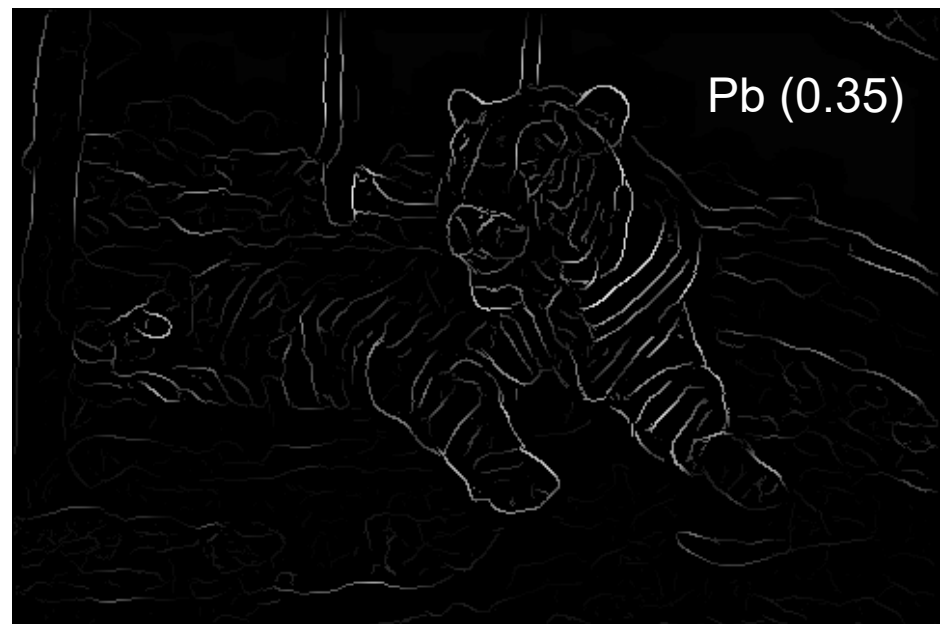
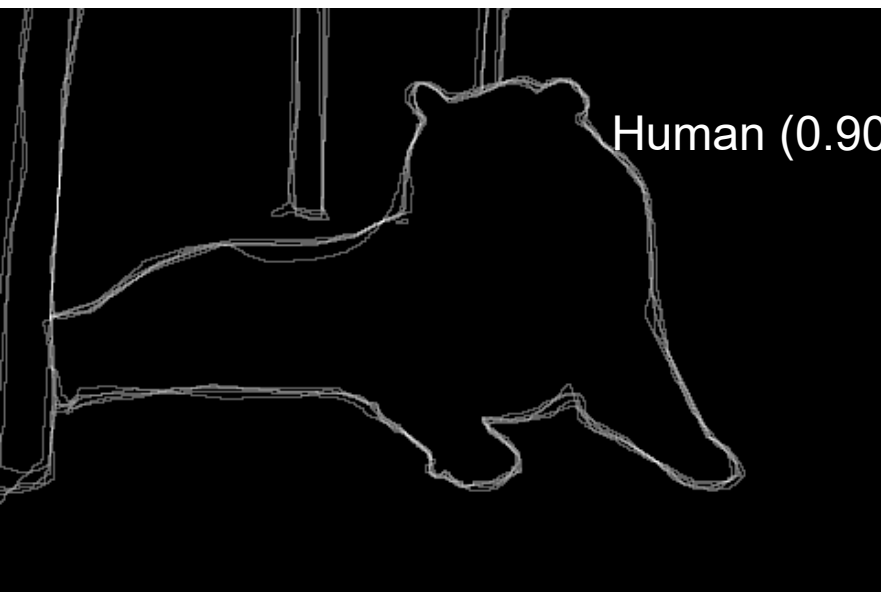
Results



Results







For more:

<http://www.eecs.berkeley.edu/Research/Projects/CS/vision/bsds/bench/html/108082-color.html>

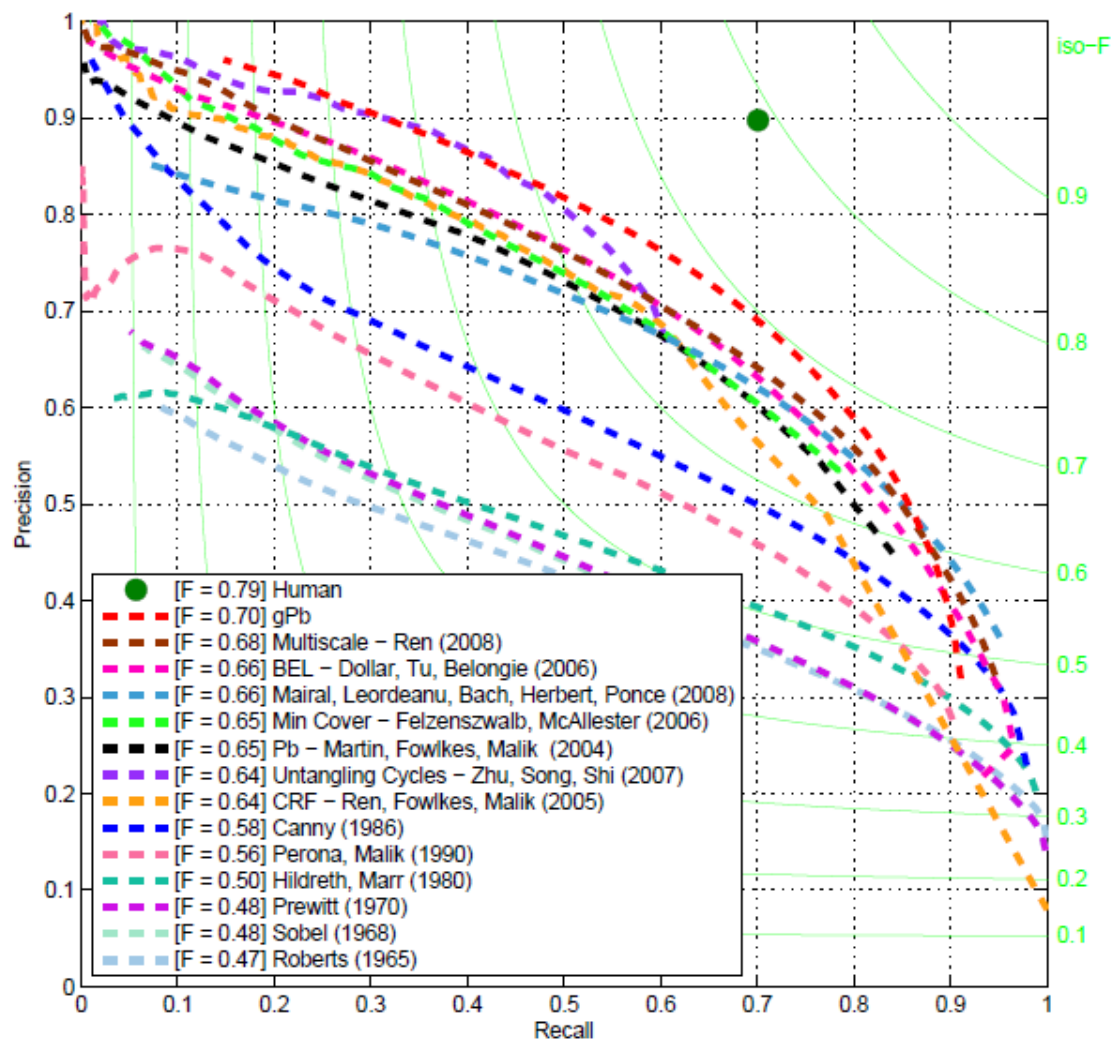


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45 years of boundary detection

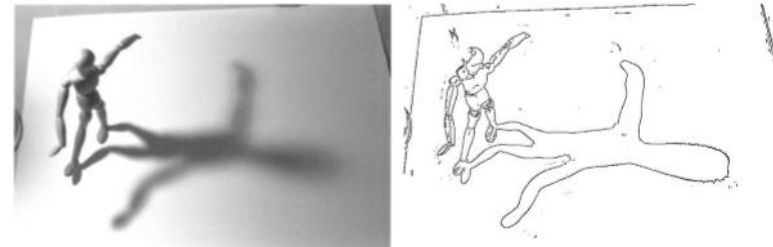
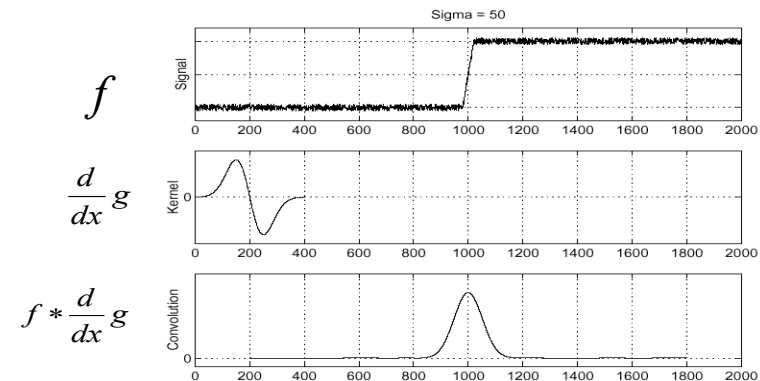


State of edge detection

- Local edge detection works well
 - ‘False positives’ from illumination and texture edges (depends on our application).
- Some methods to take into account longer contours
- Modern methods that actually “learn” from data.
- Poor use of object and high-level information.

Summary: Edges primer

- Edge detection to identify visual change in image
- Derivative of Gaussian and linear combination of convolutions
- What is an edge?
What is a good edge?



Canny edge detector

- Probably the most widely used edge detector in computer vision.
- Theoretical model: step-edges corrupted by additive Gaussian noise.
- Canny showed that first derivative of Gaussian closely approximates the operator that optimizes the product of *signal-to-noise ratio* and localization.

J. Canny, [**A Computational Approach To Edge Detection**](#), IEEE Trans. Pattern Analysis and Machine Intelligence, 8:679-714, 1986.

22,000 citations!

Demonstrator Image

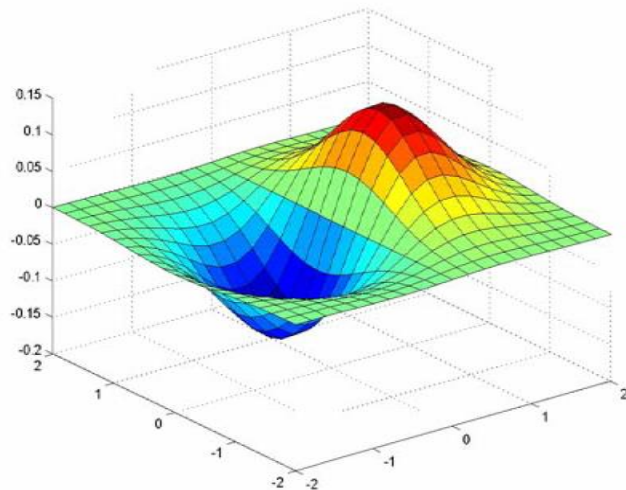
`rgb2gray('img.png')`



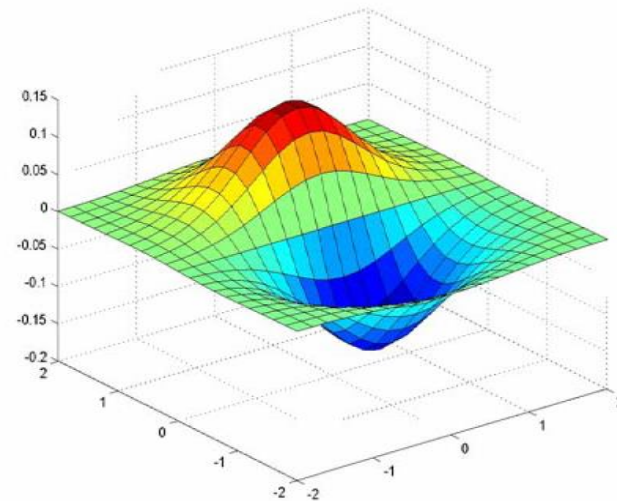
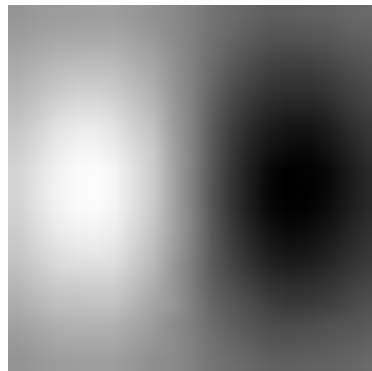
Canny edge detector

1. Filter image with x, y derivatives of Gaussian

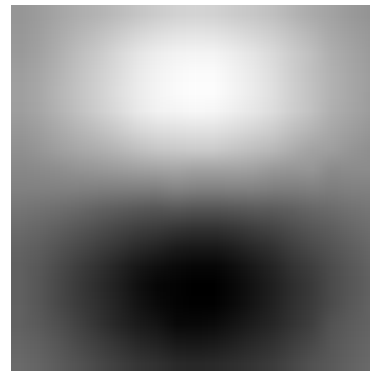
Derivative of Gaussian filter



x-direction



y-direction



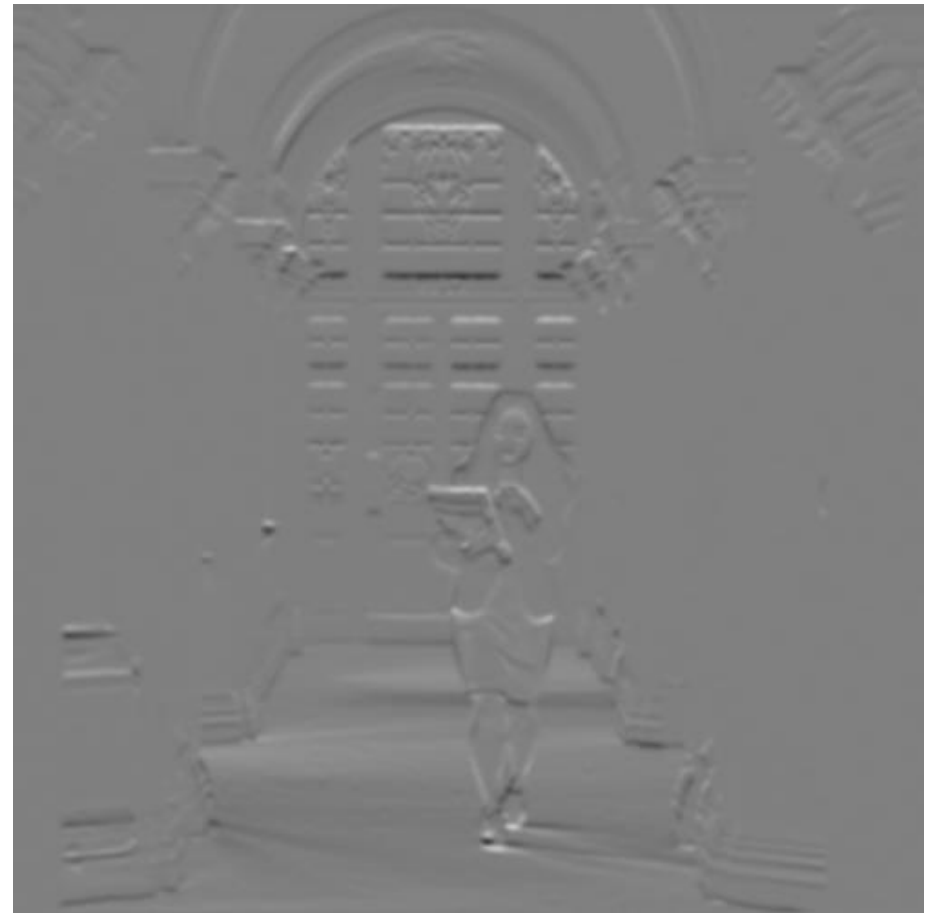
Compute Gradients



X Derivative of Gaussian



Y Derivative of Gaussian



(x2 + 0.5 for visualization)

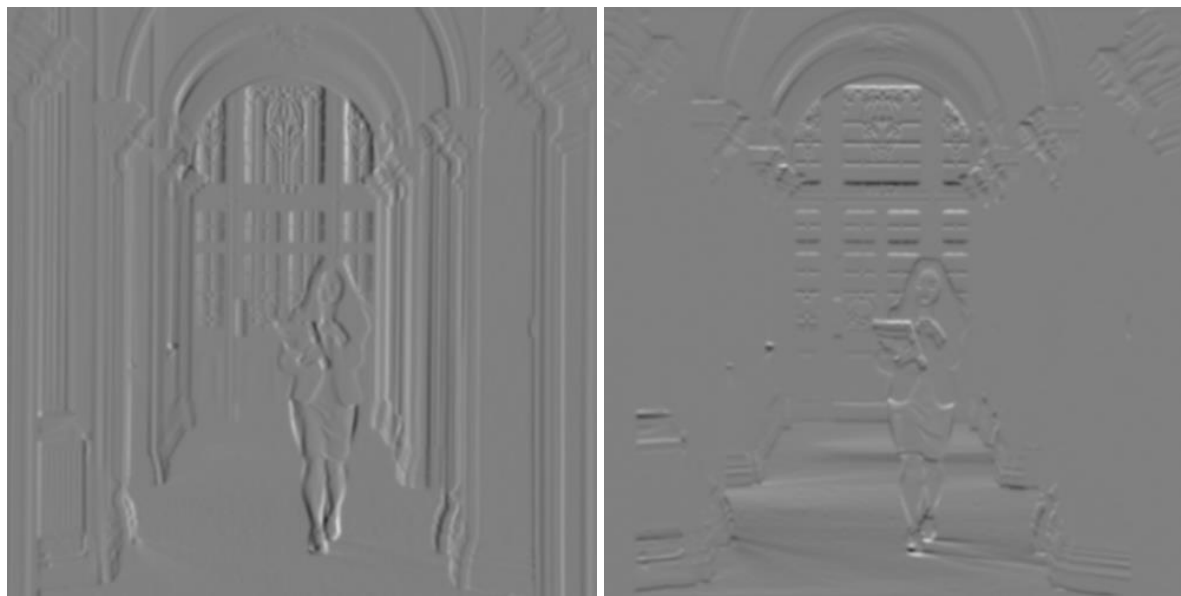
Canny edge detector

1. Filter image with x, y derivatives of Gaussian
2. Find magnitude and orientation of gradient

Compute Gradient Magnitude



$\text{sqrt}(\text{XDerivOfGaussian} .^2 + \text{YDerivOfGaussian} .^2)$ = gradient magnitude

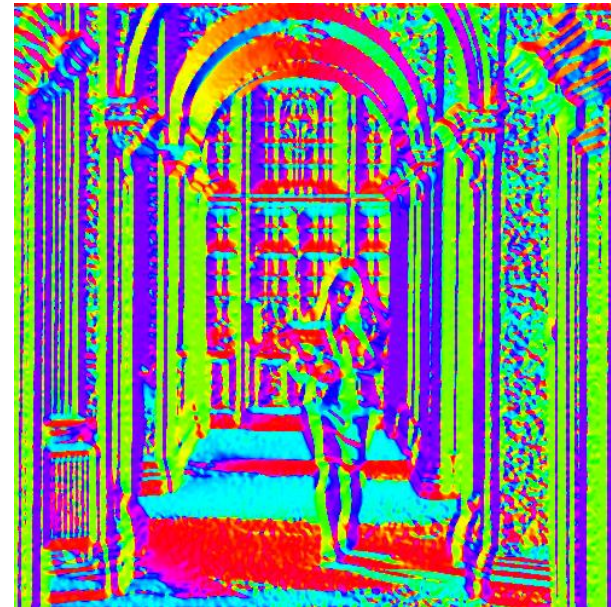
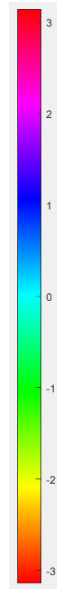


(x4 for visualization)

Compute Gradient Orientation

- Threshold magnitude at minimum level
- Get orientation via $\theta = \text{atan2}(g_y, g_x)$

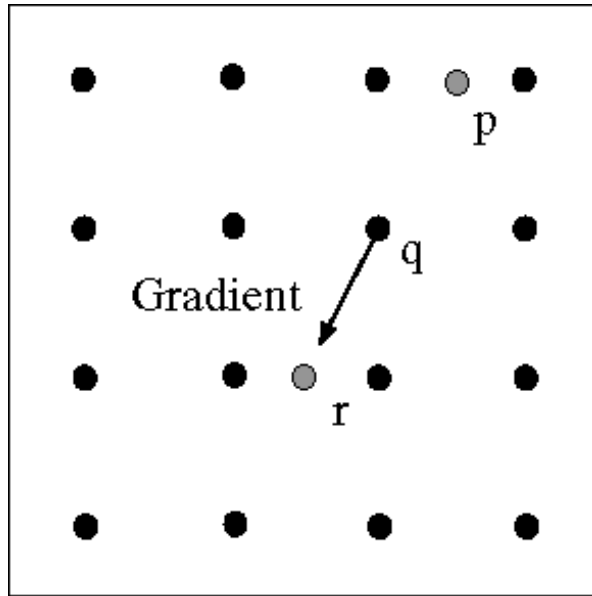
Unthresholded:



Canny edge detector

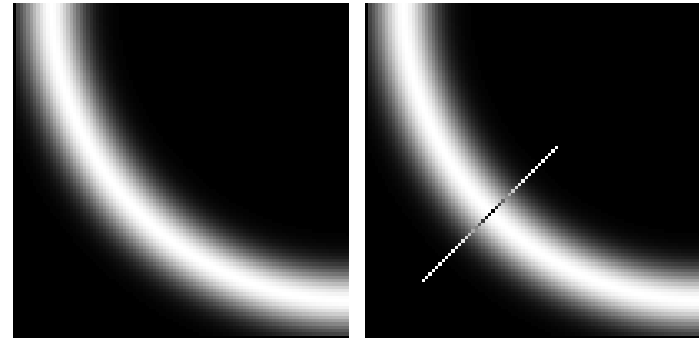
1. Filter image with x, y derivatives of Gaussian
2. Find magnitude and orientation of gradient
3. Non-maximum suppression:
 - Thin multi-pixel wide “ridges” to single pixel width

Non-maximum suppression for each orientation

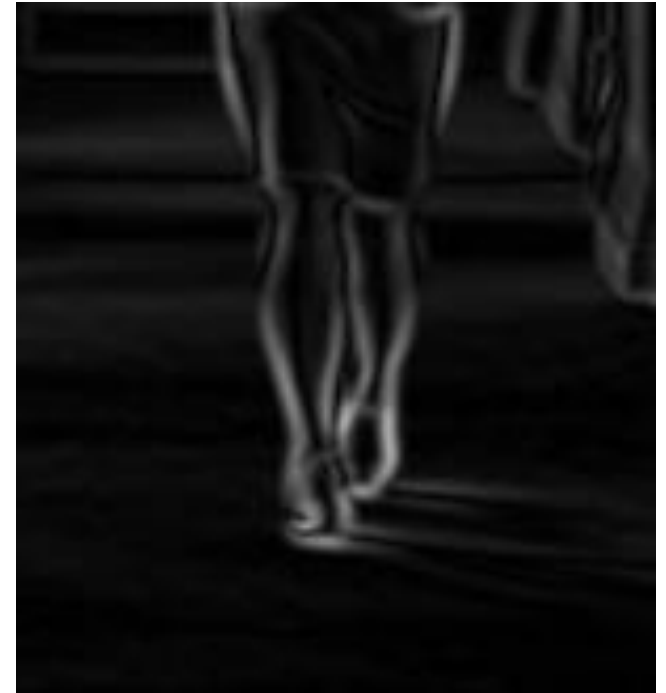


At pixel q :
We have a maximum if the value is larger than those at both p and at r .

Interpolate along gradient direction to get these values.

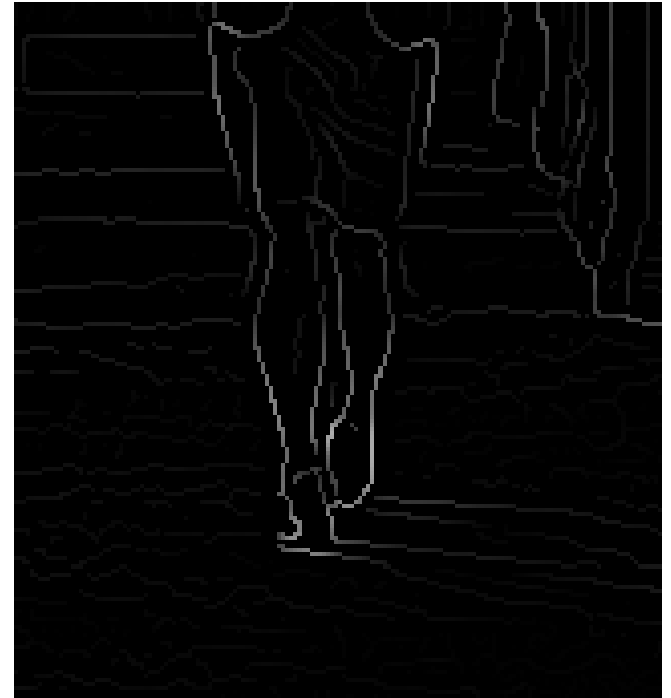


Before Non-max Suppression



Gradient magnitude (x4 for visualization)

After non-max suppression



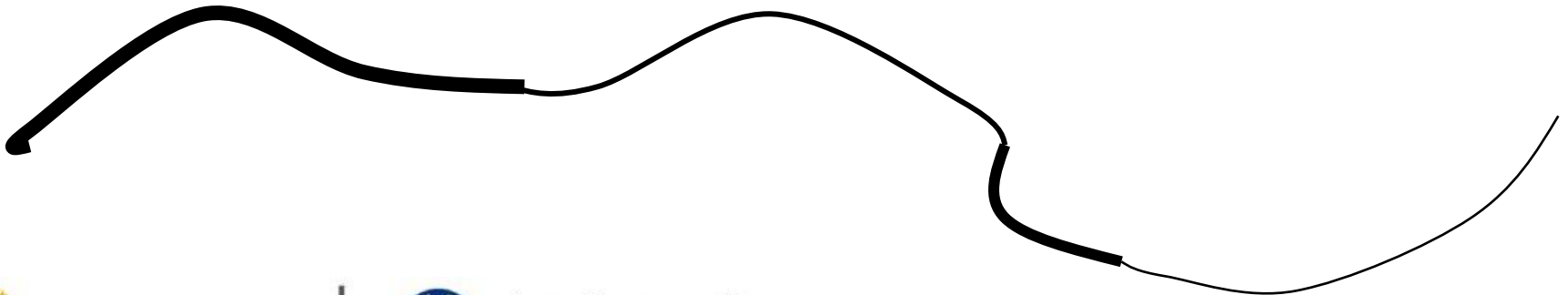
Gradient magnitude (x4 for visualization)

Canny edge detector

1. Filter image with x, y derivatives of Gaussian
2. Find magnitude and orientation of gradient
3. Non-maximum suppression:
 - Thin multi-pixel wide “ridges” to single pixel width
4. ‘Hysteresis’ Thresholding

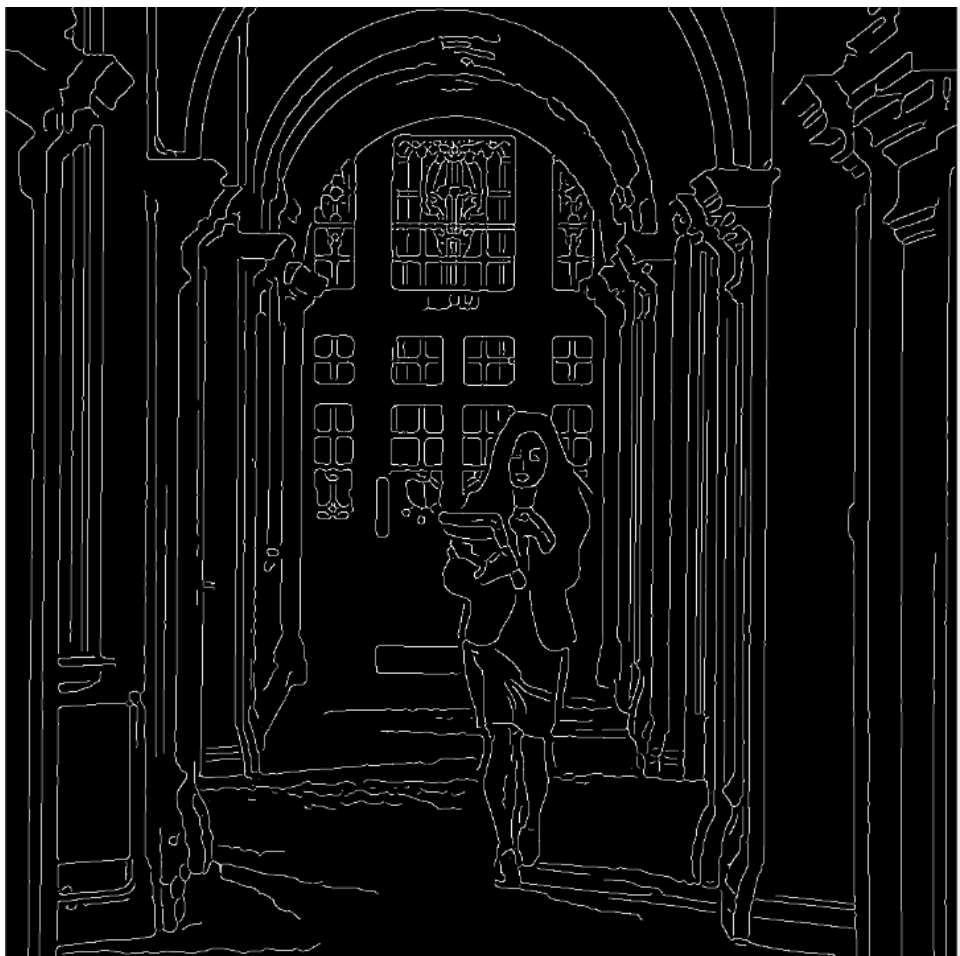
‘Hysteresis’ thresholding

- Two thresholds – high and low
- Grad. mag. > high threshold? = strong edge
- Grad. mag. < low threshold? noise
- In between = weak edge
- ‘Follow’ edges starting from strong edge pixels
- Continue them into weak edges
 - Connected components (Szeliski 3.3.4)



Final Canny Edges

$$\sigma = \sqrt{2}, t_{low} = 0.05, t_{high} = 0.1$$



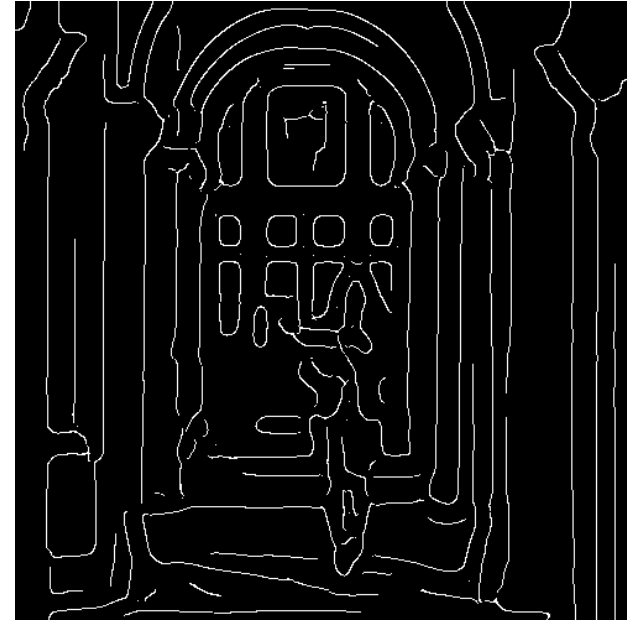
Effect of σ (Gaussian kernel spread/size)



Original



$\sigma = \sqrt{2}$



$\sigma = 4\sqrt{2}$

The choice of σ depends on desired behavior

- large σ detects large scale edges
- small σ detects fine features

Canny edge detector

1. Filter image with x, y derivatives of Gaussian
 2. Find magnitude and orientation of gradient
 3. Non-maximum suppression:
 - Thin multi-pixel wide “ridges” to single pixel width
 4. ‘Hysteresis’ Thresholding:
 - Define two thresholds: low and high
 - Use the high threshold to start edge curves and the low threshold to continue them
 - ‘Follow’ edges starting from strong edge pixels
 - Connected components (Szeliski 3.3.4)
- **MATLAB: `edge(image, 'canny')`**

PUSHING THE BOUNDARIES OF BOUNDARY DETECTION USING DEEP LEARNING

Iasonas Kokkinos

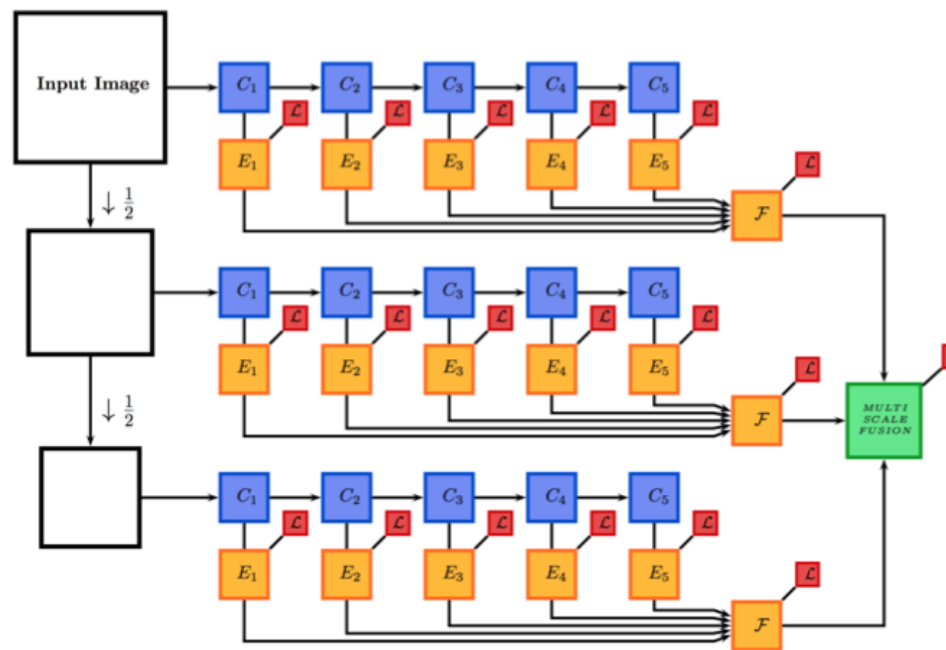


Image Pyramid



Tied CNN outputs



Scale fusion

Final outputs

Photo-Sketching: Inferring Contour Drawings from Images

WACV 2019

Mengtian Li¹

Zhe Lin²

Radomír Měch²

Ersin Yumer³

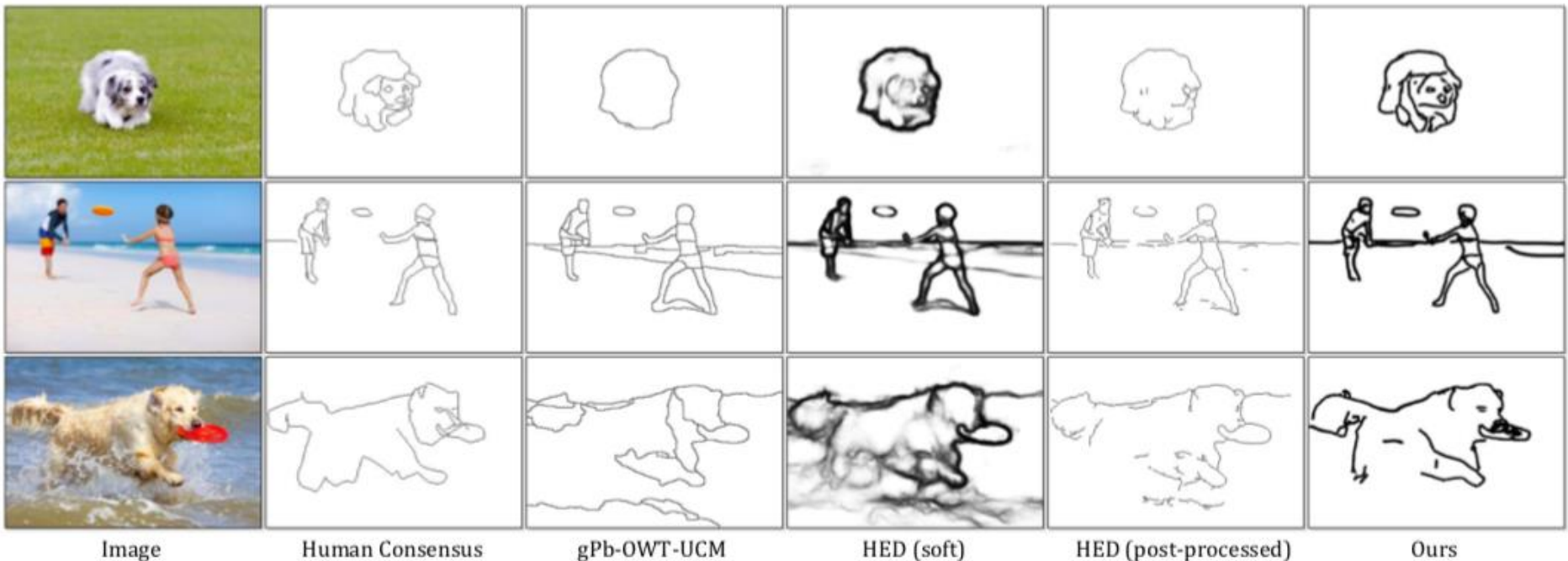
Deva Ramanan^{1,4}

¹Carnegie Mellon University

²Adobe Research

³Uber ATG

⁴Argo AI



Uses fairly advanced deep net technique (GANs), which we'll discuss only later in the course.